AIKTA ARYA

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Employment

July 2026 - Present — **Visiting Assistant Professor,** CSE Department, IIT Dharwad.

August 2025 - June 2026 — **Postdoctoral Researcher,** Korea Advanced Institute of Science & Technology (KAIST), Daejeon, South Korea.

Education

Aug 2020 – Apr 2025 — **Ph.D. CSE, Indian Institute of Technology Roorkee, India**
Addressing Challenges in Signed Networks: From Synthetic Generation to Predictive Modeling.
CGPA : 8.857/10 (Coursework)

July 2015 – June 2017 — **M.Tech. CSE, National Institute of Technology Kurukshetra, India**
CGPA : 9.2258/10 ***[Gold Medal]***

Sep 2009– Aug 2013 — **B.Tech. CSE, Uttar Pradesh Technical University, India**
Percentage : 76.06%

Research Interest

My research interests lie in modeling and analyzing networked systems, social network models and the generation of synthetic networks for downstream applications such as triangle counting, link prediction, and network modeling .

Research Publications

Journal Articles

1. A. Arya, P. Pandey, N. Ganguly, and T. Derr, "A survey on signed network reconstruction modeling and its applications," ***ACM Transactions on Knowledge Discovery from Data***, 2026.
2. A. Arya and P. K. Pandey, "Sissrm: Sequentially induced signed subnetwork reconstruction model for generating realistic synthetic signed networks," ***IEEE Transactions on Computational Social Systems***, pp. 1–11, 2024. DOI: 10.1109/TCSS.2024.3392613.
3. A. Arya, P. K. Pandey, and A. Saxena, "Balanced and unbalanced triangle count in signed networks," ***IEEE Transactions on Knowledge and Data Engineering***, 2023. DOI: 10.1109/TKDE.2023.3272657.
4. P. K. Pandey, A. Arya, and A. Saxena, "X-distribution: Retraceable power-law exponent of complex networks," ***ACM Transactions on Knowledge Discovery from Data***, 2023. DOI: 10.1145/3639413.
5. A. Arya and P. K. Pandey, "Structural reconstruction of signed social networks," ***IEEE Transactions on Computational Social Systems***, 2022. DOI: 10.1109/TCSS.2022.3186017.

Conference Proceedings

1. A. Arya, P. K. Pandey, and T. Derr, "Signed network dataset repository: Extracting signed relations from social networks," in ***Proceedings of the IEEE International Conference on Communications***, 2025.
2. A. Arya, "Realistic synthetic signed network generation and analysis," in ***Proceedings of the 33rd ACM International Conference on Information and Knowledge Management***, 2024, pp. 5439–5442.
3. A. Arya and A. K. Singh, "An efficient message forwarding in delay tolerant networks using power of crowd," in *2017 8th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, IEEE, 2017, pp. 1–5.
4. A. Arya and A. K. Singh, "An efficient routing protocol for dtn," in *International Conference on Advanced Informatics for Computing Research*, Springer, 2017, pp. 271–282.

Books and Chapters

1. A. Arya, P. K. Pandey, and A. Saxena, "Node classification using deep learning in social networks," in *Deep Learning for Social Media Data Analytics*, Springer, 2022, pp. 3–26.

Employment History

Dec 2021 - Oct 2023 — **Senior Research Fellow,** in the project "Immunity of Large Networks", funded by DST, India at Dept. of CSE, IIT Roorkee.

Dec 2019 - Dec 2021 — **Junior Research Fellow,** in the project "Immunity of Large Networks", funded by DST, India at Dept. of CSE, IIT Roorkee.

July 2017- May 2019 — **Assistant Professor,** GLA University, India.

Awards, Grants and Fellowships

Awards and Achievements

2025 — **Second Position**, in Institute Research Scholar day 2025 presentation session at IIT Roorkee, INDIA.

First Position, in Research Scholar day 2025 talk session at Department of CSE, IIT Roorkee, INDIA.

2024 — **Third Position**, in poster presentation, at the exploreCSR 2023-2024 event, sponsored by Google Research.

2023 – 2025 — **MHRD Fellowship**, at IIT Roorkee, INDIA.

2019 – 2023 — **Research Fellowship**, in a project "Immunity of Large Networks", by DST, INDIA.

2018 — **Gold Medal**, at NIT Kurukshetra, for being the best graduating student in MTech Computer Engineering.

2015 – 2017 — **MHRD Fellowship**, at NIT Kurukshetra, INDIA.

Grants and Funding

2024 — Awarded with the CIKM 2024 SIGWEB Travel Grant for attending "CIKM 2024: 33rd ACM International Conference on Information and Knowledge Management, in Boise, Idaho, USA on October 21-25, 2024." **Amount of the travel award: $1,666**

Awarded with the ICWSM 2024 Virtual Attendance Scholarship Award for participating in "ICWSM 2024: The International AAAI Conference on Web and Social Media conference, June 3-6, 2024".

Awards, Grants and Fellowships (continued)

2023 — Awarded with the SIGWEB Student Travel Grant for "The WSDM 2024: The Seventeenth ACM International Conference on Web Search and Data Mining conference, March 4-8, 2024".

Awarded with the Student Travel Grant for attending "6th International Conference on Data Science and Management of Data (CODS-COMAD)" organised by IIT Bombay.

2022 — Awarded with the Student Travel Grant for attending "Indian Symposium on Machine Learning (IndoML – 2022)" hosted by IIT Gandhinagar.

Professional Activities

2025 — Organizing Committee Member of the workshop "Building Tomorrow: Security in the Age of AI, Quantum, and 6G" jointly organized by Department of CSE, IIT Tirupati and IIT Roorkee sponsored by SPARC, India.

Program Committee Member in ICWSM 2025.

Program Committee Member in ICC 2025.

2024–present — Technical Reviewer in ACM transactions on knowledge discovery from data.

2024 — Organizing Committee Member of the workshop "Securing the Future: AI, Privacy and Zero Touch Network" organized by Department of CSE, IIT Roorkee, sponsored by SPARC, India.

Program Committee Member in ICML 2024.

Program Committee Member in NeurIPS 2024.

Program Committee Member in ICWSM 2024.

Program Committee Member in WSDM 2024.

2023 – 2024 — Elected as DAPC, Ph.D. student representative in Computer Science and Engineering Department at IIT Roorkee.

2023 — Student Organizing Committee Member in IEEE ELEXCOM 2023.

2022–present — Technical Reviewer in IEEE Transactions on Network Science and Engineering.

2021–present — Technical Reviewer in IEEE Transactions on Computational Social Systems.

2020–present — Technical Reviewer in IEEE Transactions on Knowledge and Data Engineering.

2016 - 2017 — Student Placement coordinator of M-Tech Computer Engineering, at NIT Kurukshetra, India.

Technical skills

Programming — MATLAB, PYTHON, JAVA, C.

Other — MSOffice, Latex.

References

1 **Prof. Pradumn Kumar Pandey**, Associate Professor, Department of CSE, IIT Roorkee, India. (pradumn.pandey@cs.iitr.ac.in)

2 **Prof. Ranveer Singh**, Associate Professor, Department of CSE, IIT Indore, India. (ranveer@iiti.ac.in)

3 **Prof. Tyler Derr**, Assistant Professor, Department of Computer Science, Data Science Institute, Vanderbilt University, Nashville, TN, USA. (tyler.derr@vanderbilt.edu)